“SMARTCINNAMON”: A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

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DECLARATION

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The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of the Supervisor:
(Mrs. Sanjeevi Chandrasiri)

Date:

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ABSTRACTION

Post-Harvest Ceylon Cinnamon Warehouse Quality Guard

Ceylon cinnamon is a premium export crop for Sri Lanka; however, its post-harvest quality is often affected by inadequate warehouse monitoring. Traditional methods mainly track temperature and humidity, which are not sufficient to prevent mold growth, microbial contamination, and overall quality degradation. This study proposes the Post-Harvest Cinnamon Quality Guard, an IoT and AI-based monitoring system designed to improve storage conditions.

The system uses multi-sensor arrays installed at warehouse locations to continuously measure environmental parameters including temperature, relative humidity, CO_2, VOCs, air quality, light intensity, motion and moisture. Sensor data is transmitted to a cloud-based mobile dashboard in real time, where machine learning models compare readings with predefined baseline values.

When abnormal conditions are detected, the system generates real-time alerts and provides corrective actions such as improving ventilation, activating dehumidifiers, or reducing light exposure. The expected outcomes include reduced mold development, minimized post-harvest losses, and improved consistency in cinnamon quality.

In conclusion, the Post-Harvest Cinnamon Quality Guard provides a cost-effective, proactive, and data-driven solution for cinnamon storage management, with the potential to enhance export quality and support the cinnamon industry.

Keywords - Cinnamon quality, IoT monitoring, post-harvest management, machine learning, environmental monitoring

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Ms. Sanjeevi Chandrasiri, and co-supervisor, Ms. Shalini Rupasinghe, for their invaluable guidance, continuous support, and encouragement throughout this research project. Their insights and expertise in IoT systems, environmental monitoring, and machine learning greatly contributed to shaping the direction and quality of this work.

I am also deeply grateful for their constructive feedback and technical guidance, particularly in system design, sensor integration, and the development of machine learning models used in this study.

Special thanks are extended to the Cinnamon Research Department, Sri Lanka, for providing the necessary support, domain knowledge, and resources to carry out this research successfully. Their cooperation was essential in understanding real-world challenges related to post-harvest cinnamon storage.

I would also like to express my sincere appreciation to the external supervisor, Mr. Chinthaka Vidanapathirana, for his valuable guidance and industry expertise, which helped improve the practical relevance of this research.

I also wish to thank my project team members B.M. Dilshan, Nilan Hansajith, Divana Vihani, Rushani Randeni for their collaboration, dedication, and shared effort in developing the overall SmartCinnamon system. Their teamwork played a key role in the successful completion of this project.

Finally, I would like to express my heartfelt gratitude to my family and friends for their continuous support, patience, and encouragement throughout my academic journey.

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1. INTRODUCTION

1.1 General Introduction

Ceylon cinnamon is one of the most valuable agricultural exports in Sri Lanka, recognized globally for its unique aroma, flavor, and high economic value. Maintaining this quality during the post-harvest stage is essential, as any deterioration directly affects export standards and market competitiveness. After harvesting, cinnamon is stored in warehouse environments where it becomes highly sensitive to environmental conditions. Therefore, proper monitoring and control of storage conditions are critical to preserving its quality [1].

Traditional storage monitoring systems mainly focus on temperature and relative humidity. Although these parameters are important, they are not sufficient to ensure optimal storage conditions. Cinnamon quality is influenced by multiple environmental factors, including temperature, relative humidity, moisture content, air quality, carbon dioxide (CO_2) levels, volatile organic compounds (VOC), light intensity, and motion within the storage environment. The absence of comprehensive monitoring systems often leads to mold growth, aroma loss, contamination, and physical damage, which are typically detected only after the damage has already occurred [2], [3].

Each environmental parameter has a defined safe range, and deviations from these ranges can directly impact cinnamon quality. For example, the safe temperature range is approximately 25°C to 30°C, where deviations can result in moisture imbalance or loss of essential oils. Similarly, relative humidity should be maintained between 55% and 65%, as higher levels promote mold growth while lower levels cause brittleness in cinnamon quills. Moisture content within cinnamon should remain between 10% and 14%, since excessive moisture can lead to internal rotting and reduced shelf life [4].

Air quality is another critical factor that significantly affects storage conditions. Elevated carbon dioxide (CO_2) levels above the safe range of 400–800 ppm indicate poor ventilation and stagnant air, which can encourage microbial growth and mold formation. In addition, high concentrations of volatile organic compounds (VOC) can introduce chemical odors that degrade the natural aroma of cinnamon. These factors highlight the importance of maintaining proper air circulation and ventilation within storage environments [5].

Light intensity and motion also play an important role in cinnamon storage. Excessive light exposure, particularly above 100 lux, can cause fading of the natural golden color of cinnamon, reducing its visual quality. Meanwhile, unexpected motion detected within the storage area may indicate pest activity, human interference, or potential security risks. Such conditions require immediate attention to prevent further damage or loss [6].

A major limitation of existing storage monitoring approaches is their reactive nature. Most systems either record environmental data or generate alerts only after critical thresholds have been exceeded. This delays corrective actions and increases the risk of irreversible quality degradation. Therefore, a proactive monitoring system is required to continuously track environmental parameters, detect risks early, and provide appropriate corrective actions [7].

To address these challenges, this research introduces the Post-Harvest Cinnamon Quality Guard, an IoT and machine learning-based system designed for real-time environmental monitoring and intelligent risk detection. The system integrates multiple sensors to continuously collect data related to temperature, humidity, moisture, air quality (CO_2 and VOC), light intensity, and motion. These sensor readings are compared with predefined safe ranges, and a machine learning model is used to evaluate deviations and determine risk levels [8].

Based on this analysis, the system generates real-time alerts and provides corrective actions such as improving ventilation, activating dehumidifiers, reducing light exposure, or inspecting the warehouse. This approach enables early intervention, reduces post-harvest losses, and helps maintain consistent cinnamon quality during storage.

In conclusion, the proposed system offers a comprehensive, proactive, and data-driven solution for cinnamon storage management. By integrating multi-parameter monitoring with intelligent analysis and action recommendations, it supports efficient post-harvest handling and enhances the overall sustainability and competitiveness of Sri Lanka's cinnamon industry.

1.2 Background Literature

1.2.1 Environmental Monitoring and Multi-Sensor Data Collection

Post-harvest storage monitoring plays a crucial role in preserving the quality of Ceylon cinnamon. During storage, multiple environmental parameters influence product stability, and failure to control these conditions can lead to quality degradation such as mold formation, aroma loss, and structural damage. Therefore, continuous and accurate monitoring of environmental conditions is essential.

Traditional monitoring systems mainly focus on limited parameters such as temperature and humidity, using manual instruments. However, these approaches are insufficient for modern storage environments as they do not provide real-time monitoring and fail to capture complex environmental interactions [1]. As a result, advanced monitoring systems now adopt a multi-sensor approach to improve accuracy and reliability.

The cinnamon warehouse system integrates multiple IoT sensors to monitor key environmental parameters simultaneously. These parameters include:

- Temperature (25°C – 30°C) – Maintains chemical stability and prevents essential oil loss
- Humidity (55% – 65%) – Controls mold growth and prevents brittleness
- Moisture (10% – 14%) – Prevents internal rotting and quality degradation
- Air Quality (CO_2: 400–800 ppm) – Ensures proper ventilation and prevents contamination
- VOC Levels (Low range) – Protects natural aroma and chemical composition
- Light Intensity (Low exposure) – Prevents discoloration
- Motion Detection (No disturbance) – Detects pests or unauthorized access
- Airflow / Environment Stability – Maintains balanced internal conditions

Each of these parameters is continuously monitored using sensors such as DHT22 (temperature and humidity), MQ135 (air quality and VOC), PIR sensors (motion), and LDR sensors (light intensity). The collected data provides a comprehensive understanding of the storage environment.

Unlike traditional systems, this multi-parameter approach enables the system to analyze the combined effect of environmental conditions. For example, high humidity combined with poor air quality significantly increases the risk of mold formation. Therefore, integrating multiple sensors improves accuracy and enables early detection of potential risks [9].

Figure 1: Multi-Sensor Environmental Parameters in Cinnamon Warehouse System

This figure 1 illustrates the key environmental parameters monitored in the cinnamon warehouse monitoring component of the SmartCinnamon system. It includes temperature, humidity, moisture, CO_2 levels, VOC, light intensity, motion detection, and air quality, which are essential for maintaining optimal storage conditions and preventing quality degradation.

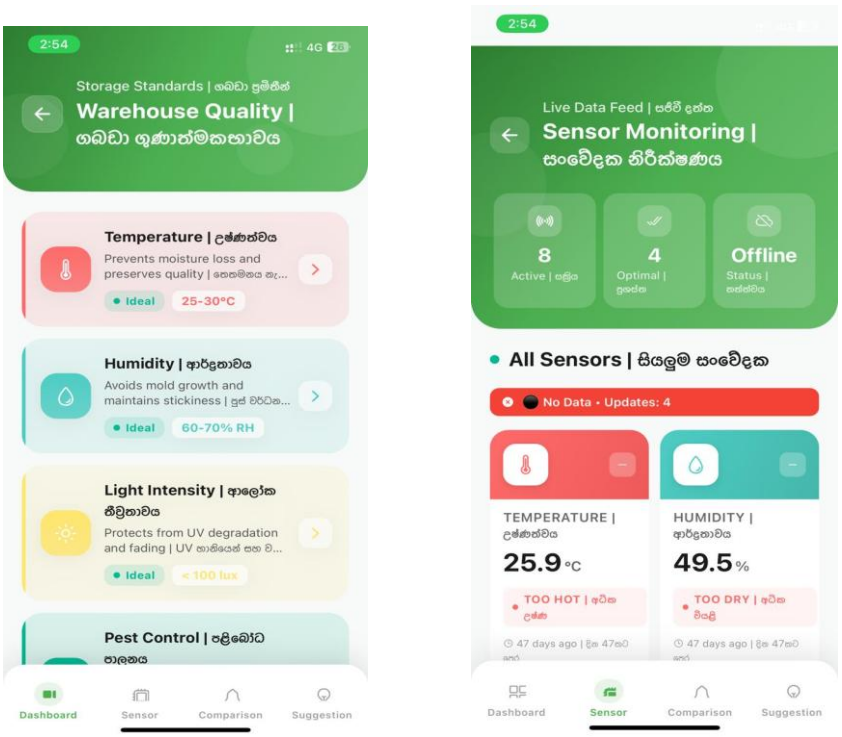

Figure 2: Real-Time Environmental Monitoring Dashboard of Mobile App in Warehouse Component

Figure 2 illustrates the mobile application dashboard used in the warehouse monitoring component of the SmartCinnamon system. The interface provides real-time visualization of environmental parameters including temperature, humidity, moisture, air quality (CO_2), VOC levels, light intensity, and motion detection. This enables users to monitor storage conditions effectively and respond to abnormal situations promptly.

The integration of these sensors with IoT technology allows real-time data transmission to a cloud platform such as Firebase. This enables remote monitoring through a mobile application, where users can view environmental conditions, track parameter changes, and respond to abnormal situations instantly.

Furthermore, continuous data collection allows the system to build historical datasets, which can be used for further analysis and prediction. This enhances the capability of the system to not only monitor but also support intelligent decision-making.

Overall, the use of multi-sensor data collection in the SmartCinnamon system provides a reliable and efficient solution for monitoring storage conditions. By combining multiple environmental parameters, real-time data processing, and mobile-based visualization, the system ensures improved product quality and reduced post-harvest losses.

1.2.2 Environmental Conditions and Risk Analysis

Following the multi-sensor data collection approach illustrated in Figure 1, it is essential to understand how these environmental parameters influence the storage quality of Ceylon cinnamon. Monitoring alone is not sufficient unless the collected data is properly interpreted to identify potential risks. Therefore, analyzing environmental conditions and their impact on product quality is a key aspect of modern storage systems.

Each environmental parameter plays a specific role in determining the stability of stored cinnamon. Temperature directly affects the chemical composition of the product. When temperature levels exceed the optimal range, essential oils begin to evaporate, resulting in a loss of aroma and flavor. On the other hand, low temperature conditions can affect the structural integrity of cinnamon quills, making them brittle and more susceptible to breakage.

Humidity is another critical factor that significantly influences storage conditions. High humidity levels create a favorable environment for mold growth and microbial activity, which can lead to contamination and quality degradation. In contrast, low humidity levels cause

excessive drying, reducing the flexibility and weight of cinnamon. Therefore, maintaining humidity within a controlled range is essential to preserve product quality.

Moisture content is closely related to both temperature and humidity. Excess moisture can result in internal rotting and fungal growth, while insufficient moisture affects the texture and overall value of the product. Continuous monitoring of these parameters ensures that moisture levels remain within safe limits.

In addition to these primary factors, air-related parameters such as CO_2 levels, VOC presence, and overall air quality provide deeper insight into storage conditions. Poor air circulation and gas accumulation can indicate an unstable environment, increasing the risk of contamination. Similarly, light intensity affects the visual quality of cinnamon, where excessive exposure may lead to discoloration. Motion detection further supports monitoring by identifying disturbances such as pest activity or unauthorized access.

Unlike traditional systems that evaluate each parameter independently, modern approaches focus on the combined effect of multiple environmental conditions. For example, a moderate increase in humidity alone may not be a serious issue. However, when combined with elevated temperature and poor air quality, the risk of mold formation increases significantly. This highlights the importance of multi-parameter analysis in accurately identifying potential risks.

By continuously comparing real-time sensor data with predefined safe ranges, the system can detect abnormal conditions and classify them into risk levels. This analysis forms the foundation for advanced features such as alert generation and decision support, which are discussed in subsequent sections.

1.2.3 Intelligent Risk Detection using Machine Learning

After analyzing environmental conditions and identifying associated risks, it is essential to process the collected data in an intelligent manner. Traditional monitoring systems mainly rely on fixed threshold values to detect abnormal conditions. However, this approach is limited because it evaluates each parameter separately and cannot capture the complex relationships between multiple environmental factors.

To overcome these limitations, modern monitoring systems incorporate machine learning techniques. Machine learning models can analyze large volumes of sensor data and identifying

hidden patterns that indicate potential risks. This enables early detection of abnormal conditions before visible damage occurs.

In the cinnamon warehouse monitoring component of the SmartCinnamon system, the Random Forest algorithm is used for risk detection. Random Forest is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reliability. It is particularly suitable for multi-parameter environments where several environmental factors interact simultaneously [13].

Unlike traditional systems, the machine learning model does not simply check whether a parameter exceeds a threshold. Instead, it evaluates the overall pattern of environmental data. For example, a slight increase in humidity alone may not indicate a critical issue. However, when combined with rising temperature and poor air quality, it can significantly increase the risk of mold formation. The model can recognize such patterns and classify the situation accordingly.

To further highlight the effectiveness of machine learning approaches, a comparison between traditional threshold-based systems and machine learning-based risk detection is illustrated in Figure 4.

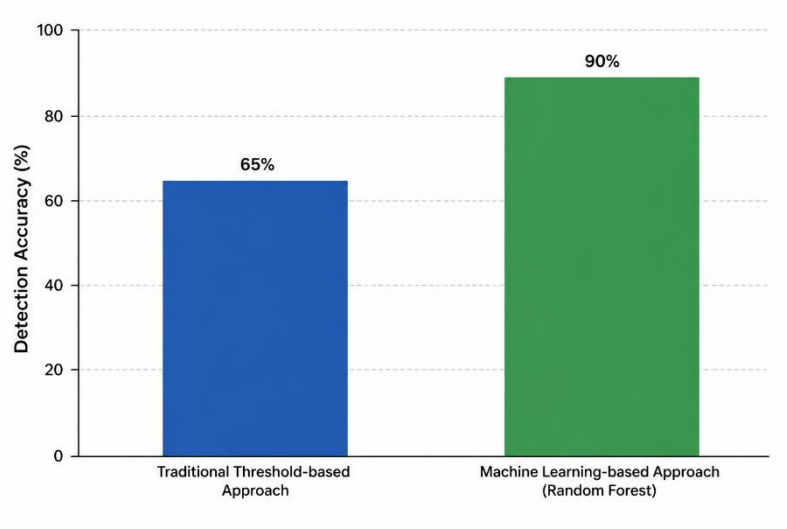

Figure 3: Comparison of Risk Detection Accuracy between Traditional and Machine Learning Approaches

The chart demonstrates that machine learning models provide higher accuracy in detecting environmental risks compared to traditional methods. This improvement is due to the ability of ML models to analyze multiple parameters simultaneously and identify complex relationships between them.

In addition, machine learning models can continuously improve by learning from historical data. As more data is collected over time, the system becomes more accurate and adaptive. This allows the system to predict potential risks in advance and support proactive decision-making.

Overall, the integration of machine learning transforms the monitoring system from a passive data collection tool into an intelligent system capable of predictive analysis and proactive risk management.

1.2.4 IoT Integration, Real-Time Alerts, and Mobile Application Interaction

Following the intelligent risk detection process described in the previous section, it is essential to ensure that the identified risks are effectively communicated to users in real time. This is achieved through the integration of Internet of Things (IoT) technology, real-time alert mechanisms, and mobile application interfaces.

The warehouse monitoring component of the SmartCinnamon system utilizes IoT-based architecture to enable continuous data collection and transmission. Sensors deployed within the storage environment capture environmental parameters such as temperature, humidity, moisture, air quality (CO_2), VOC levels, light intensity, and motion detection. These sensor readings are transmitted to a cloud-based platform, such as Firebase, where data is stored and processed in real time. IoT-based systems provide seamless connectivity and enable real-time monitoring capabilities, which are essential for modern smart environments [7].

The use of IoT technology ensures that environmental data is continuously updated and accessible remotely. This allows users to monitor warehouse conditions from any location without being physically present. Real-time data transmission also ensures that any sudden changes in environmental conditions are immediately detected and processed by the system [9].

Once the system identifies abnormal conditions or high-risk situations through machine learning analysis, it generates real-time alerts. These alerts are delivered to users through the mobile application in the form of notifications, warning messages, or visual indicators. Immediate alert generation plays a critical role in reducing response time and preventing potential damage to stored cinnamon.

In addition to alert generation, the system provides decision support by recommending appropriate corrective actions based on the detected conditions. For instance, if humidity levels exceed the safe range, the system may suggest reducing moisture levels or improving ventilation conditions. Such intelligent decision support mechanisms are widely used in smart agricultural systems to improve efficiency and reduce human intervention [6].

The mobile application serves as the primary interface for user interaction within the cinnamon warehouse system. It is developed using modern frameworks such as React Native and provides a user-friendly dashboard for monitoring environmental conditions. The application displays real-time parameter values, risk levels, and system alerts in a clear and structured manner.

As shown in Figure 2, the mobile dashboard presents all monitored environmental parameters in a centralized interface. Users can easily view current conditions, track changes over time, and respond to alerts efficiently. The integration of IoT systems with mobile applications significantly improves usability and accessibility in real-time monitoring environments [15]. Overall, the combination of IoT-based data transmission, real-time alert systems, and mobile application interfaces transforms the monitoring component into an intelligent system capable of proactive decision-making and efficient storage management.

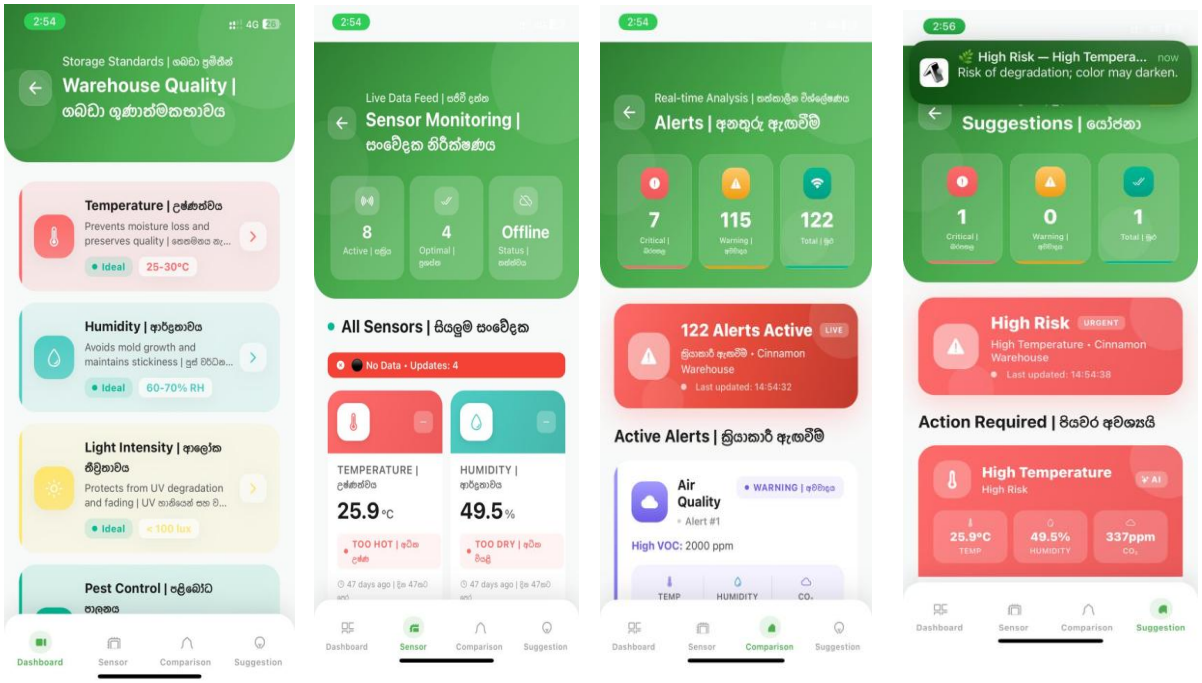

Figure 4: Warehouse Cinnamon Quality Mobile APP Integration

1.3 Research Gap

This comparative analysis highlights the key research gaps identified in existing cinnamon post-harvest storage monitoring systems. Previous studies (Research A, B, C, and D) demonstrate partial solutions for environmental monitoring and control; however, they fail to provide a comprehensive and intelligent system that addresses all critical requirements of cinnamon warehouse management.

• **Research A (Tharaka et al., 2021)** focuses mainly on temperature and relative humidity (Temp/RH) control within a hybrid cinnamon storage chamber. While it provides basic automation such as activating fans and generating alerts, the system is limited to only two environmental parameters. It does not include other essential factors such as moisture levels, CO_2 concentration, VOC detection, light intensity, motion detection, and overall air quality. These parameters are crucial for accurately identifying early-stage quality degradation. Furthermore, the system lacks machine learning–based risk prediction and does not provide decision-support recommendations, making it reactive rather than proactive.

• **Research B (Jarašūnienė et al., 2023)** emphasizes IoT-based monitoring and cloud connectivity, enabling real-time dashboards and remote accessibility. Although this system improves scalability and user accessibility, it remains generic and not specifically designed for cinnamon storage conditions. It does not incorporate cinnamon-specific parameters such as moisture content, VOC levels related to aroma preservation, or motion detection for warehouse disturbances. In addition, the system does not include intelligent risk analysis or predictive models, limiting its ability to prevent quality loss effectively.

• **Research C (Sathya et al., 2023)** introduces a multi-parameter IoT-based monitoring system with temperature, humidity, and gas detection capabilities. While this approach is more advanced than earlier systems, it still lacks several critical parameters such as moisture content, light intensity, motion detection, and comprehensive air quality monitoring. Moreover, the system relies on rule-based threshold alerts rather than machine learning models, making it reactive and unable to predict risks such as mold growth or aroma degradation in advance.

• **Research D (Feltes et al., 2023)** focuses on VOC-based analysis for distinguishing true and

false cinnamon using machine learning techniques. Although this research provides valuable insights into cinnamon composition and aroma characteristics, it does not address warehouse monitoring or environmental control. It lacks real-time IoT integration, multi-sensor environmental monitoring, and alert-based risk management, making it unsuitable as a complete storage monitoring solution.

Table I : Research Gap

Conducted By	Multi-parameter sensing	Cinnamon-specific storage baselines	ML risk prediction	Real-time alerts & actions	Warehouse-wide deployment (multi-node)
ResearchA Tharaka et al. (2021) (SLJoT)[11]	✗ (Temp/RH only)	✓ (Temp/RH limits)	✗	✓ (Bluetooth alerts; auto fan/heater)	✗ (single chamber)
ResearchB Jarašūnienė et al. (2023) (Sensors)[12]	✗	✗	✗	✗	✓
ResearchC IoT-Based Intelligent Warehouse Monitoring & Alerting (2023)[3]	✓ (Temp/RH + gas; threshold rules)	✗	✗	✓ (app/SMS threshold alerts)	✓ (multi-node prototype)
Research D Feltes et al. (2023) (Micromachines)[5]	✓ (VOCs focus)	✗	✓ (not warehouse risk)	✗	✗
Proposed system (this research)	✓ (multi parameters; most effective selected)	✓ (cinnamon baselines)	✓ (risk score + feature rules)	✓ (app/SMS + optional auto-control)	✓

Proposed System Contribution

The proposed SmartCinnamon warehouse monitoring component addresses existing limitations by monitoring eight key environmental parameters: temperature, humidity, moisture, CO_2, VOC, light, motion, and air quality. The system uses cinnamon-specific baseline ranges and applies machine learning (Random Forest) to detect and predict risks such as mold formation and aroma loss.

It provides real-time alerts and recommended actions through a mobile application, while IoT integration enables continuous monitoring and remote access. This approach ensures a proactive and efficient solution for maintaining cinnamon storage quality.

1.4 RESEARCH PROBLEM

Ceylon cinnamon, also called "true cinnamon," is one of the most valuable spices in the world, and Sri Lanka is the leading producer recognized for its unique aroma, flavor, and health benefits [1]. Cinnamon plays an important role in the Sri Lankan economy and supports many livelihoods; therefore, maintaining post-harvest quality is crucial to preserve value, satisfy customers, and protect the country's reputation in international markets. However, maintaining cinnamon quality during storage remains a significant challenge for the industry [13].

After harvest, cinnamon is typically stored in warehouses before processing, packaging, and exporting, during which it is highly sensitive to environmental conditions. Even small changes in temperature, humidity, moisture, or air quality can affect quality. High humidity and moisture encourage mold formation, while poor air quality, VOC exposure, and light intensity can reduce aroma, which is the key feature that distinguishes Ceylon cinnamon from cassia [5], [14]. Aroma loss and mold contamination reduce market value, cause financial losses, and risk Sri Lanka's premium reputation. Therefore, controlling the storage environment is essential to prevent spoilage and maintain export quality standards.

Currently, warehouse monitoring practices are basic, where most systems focus only on temperature and relative humidity or rely on manual inspections. While these parameters are important, they are not sufficient to fully protect cinnamon quality. Other critical factors include moisture levels, CO_2 concentration, VOC levels, light intensity, motion disturbances, and overall air quality. Ignoring these factors allows risks such as mold growth and aroma degradation to develop unnoticed until corrective actions are too late.

Another major limitation is that existing systems are reactive rather than proactive. Most systems only record data or generate alerts after thresholds are exceeded (for example, when humidity suddenly increases), by which time quality degradation may have already started [4], [11]. This reactive approach prevents warehouse managers from anticipating risks and taking preventive actions, leading to losses in both quality and quantity. For a high-value export product like Ceylon cinnamon, this is a serious concern affecting economic value, reputation, and sustainability [3].

To address these issues, a dedicated cinnamon-specific warehouse monitoring system is required. Such a system should monitor multiple environmental parameters in real time, rather than focusing only on temperature and humidity. In this research, the system focuses on eight key parameters: temperature, humidity, moisture, CO_2 levels, VOC levels, light intensity, motion detection, and air quality. Monitoring these parameters together provides a more complete understanding of storage conditions and helps identify potential risks early.

Furthermore, the system should apply machine learning (ML) techniques to analyze sensor data and compare it with predefined safe ranges. This allows early detection of abnormal patterns and prediction of risks such as mold formation and aroma loss before they occur [3], [5], [8]. In addition, real-time alerts and recommended actions can help users take immediate corrective measures, such as reducing humidity or controlling environmental exposure. A cloud-based dashboard can further provide real-time monitoring and historical analysis to support better decision-making [4].

Despite the importance of maintaining post-harvest cinnamon quality, existing warehouse monitoring systems are limited to basic parameters and reactive approaches. They do not support comprehensive multi-parameter monitoring, predictive risk detection, or real-time decision support.

Therefore, there is a lack of an integrated, intelligent system that can monitor key environmental factors, detect risks early, and assist users in taking preventive actions. This limitation leads to quality degradation, financial losses, and reduced competitiveness in the global market.

2. OBJECTIVES

The purpose of this research is to design and develop an intelligent and comprehensive solution for improving post-harvest cinnamon storage management through the warehouse monitoring component of the SmartCinnamon system. Based on the research problem identified earlier, this study focuses on overcoming the limitations of traditional monitoring methods and existing warehouse systems, which are often limited to basic environmental tracking and reactive responses.

The proposed system aims to create a smart monitoring environment that not only measures environmental conditions but also analyzes, predicts, and responds to potential risks affecting cinnamon quality. By integrating multi-sensor data collection, machine learning techniques, and real-time communication, the system is expected to enhance decision-making, reduce post-harvest losses, and maintain the premium quality of Ceylon cinnamon in storage environments.

Since warehouse conditions are influenced by multiple environmental factors that continuously change over time, the system must be capable of providing accurate, real-time, adaptive, and user-friendly monitoring capabilities. Therefore, the objectives of this research are structured into a main objective and several specific objectives, each focusing on a critical component of the proposed system.

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate an intelligent warehouse monitoring component for the SmartCinnamon system that improves post-harvest storage management of Ceylon cinnamon through the integration of multi-sensor monitoring, machine learning–based risk analysis, real-time alert mechanisms, and mobile application interaction.

This main objective reflects the overall intention of the study. It is not limited to simply monitoring environmental conditions but instead focuses on developing a comprehensive system that can understand, analyze, and respond to storage risks in an effective and timely manner. The system is expected to support both basic monitoring and advanced analytical capabilities by continuously tracking key environmental parameters such as temperature, humidity, moisture, CO_2 levels, VOC levels, light intensity, motion detection, and overall air

quality.

The inclusion of multi-sensor monitoring in this objective is important because it enables the system to capture a complete and accurate representation of warehouse conditions rather than relying on limited parameters. Machine learning integration is included to enhance the system's ability to analyze complex environmental patterns and predict potential risks such as mold formation and aroma degradation before they occur.

Furthermore, the use of real-time alert mechanisms ensures that users are immediately notified when abnormal conditions are detected, allowing for faster response and prevention of damage. The mobile application component is designed to provide a user-friendly interface for monitoring environmental data, viewing risk levels, and receiving recommended actions, thereby improving usability and decision-making efficiency.
Overall, this main objective represents the development of a complete and intelligent warehouse monitoring component that combines IoT technologies, data analytics, and user interaction to enhance storage quality, reduce post-harvest losses, and maintain the premium standard of Ceylon cinnamon in global markets.

2.2 Specific Objectives

To achieve the main objective of this research, several specific objectives have been identified. Each specific objective focuses on a key functional and technical aspect of the SmartCinnamon warehouse monitoring component.

2.2.1 To develop a multi-sensor environmental monitoring system for warehouse conditions

One of the primary objectives of this research is to design and implement a multi-sensor monitoring system that continuously measures key environmental parameters within the cinnamon warehouse. These parameters include temperature, humidity, moisture, CO_2 levels, VOC levels, light intensity, motion detection, and overall air quality.

The purpose of this objective is to ensure that the system captures a complete and accurate representation of the storage environment. Rather than relying on limited parameters such as temperature and humidity, the system integrates multiple sensors to provide a more

comprehensive understanding of environmental conditions. This objective is important because maintaining stable environmental conditions is essential to prevent issues such as mold formation, aroma degradation, and contamination.

2.2.2 To establish cinnamon-specific environmental baseline ranges for accurate monitoring

Another key objective is to define and implement safe baseline ranges for each environmental parameter based on optimal storage conditions for Ceylon cinnamon. These baseline values are used to evaluate real-time sensor data and identify deviations that may indicate potential risks.

The importance of this objective lies in ensuring that the system is specifically tailored to cinnamon storage requirements rather than using general agricultural standards. By applying accurate and context-specific thresholds, the system can provide more reliable monitoring and early detection of unfavorable conditions. This objective supports precise decision-making and improves the effectiveness of the monitoring process.

2.2.3 To implement machine learning for predictive risk analysis

This research also aims to integrate machine learning techniques, specifically the Random Forest algorithm, to analyze environmental data and predict potential risks. The model will process real-time sensor data and identify patterns that may lead to issues such as mold growth, aroma loss, or environmental instability.

The purpose of this objective is to move beyond traditional rule-based monitoring systems and introduce predictive capabilities. Instead of reacting only after a problem occurs, the system can anticipate risks in advance and provide early warnings. This objective is essential for enabling proactive warehouse management and minimizing post-harvest losses.

2.2.4 To design a risk classification and real-time alert system

Another important objective is to develop a mechanism that classifies environmental conditions into different risk levels such as low, medium, and high. Based on this classification, the system generates real-time alerts and notifications to inform users of abnormal conditions.

The reason for this objective is to improve response time and ensure that users are immediately aware of potential risks. By providing timely alerts, the system allows users to take corrective actions before the situation becomes critical. This objective enhances system efficiency and reduces the impact of environmental variations.

2.2.5 To provide decision support and mobile-based monitoring

This study also aims to provide decision support by recommending appropriate actions based on detected environmental conditions. In addition, a mobile application will be developed to display real-time data, alerts, and recommendations in a user-friendly manner.

The purpose of this objective is to improve usability and support informed decision-making. Users can monitor warehouse conditions remotely and take necessary actions efficiently without requiring expert knowledge.

2.2.6 To integrate IoT and cloud technologies for continuous monitoring and evaluation

The final objective is to integrate IoT-based sensors with cloud platforms such as Firebase to enable continuous data collection, storage, and remote access. The system will also be evaluated to measure its effectiveness in improving storage quality and reducing post-harvest losses.

This objective ensures that the system is scalable, reliable, and capable of supporting real-time monitoring in practical warehouse environments. It also provides evidence of the system's effectiveness in maintaining cinnamon quality.

3.SYSTEM METHODOLOGY

SMARTCINNAMON is an integrated AI–IoT platform for the cinnamon value chain. Field and warehouse devices stream data to Firebase cloud services; machine-learning models turn that data into real-time recommendations, risk alerts, and dashboards delivered through a mobile application.

At the center is the user's mobile device connected to cloud services. IoT sensors in fields and warehouses send readings to Firebase (via real-time communication), where data is stored and managed. Cloud-based machine learning models analyze the incoming data continuously and return insights to dashboards and alerting channels. The platform combines several modules that work together to improve yield, protect post-harvest quality, and support decision-making.

Real-time Soil & Fertilizer Intelligence - Sensors measure soil moisture and environmental conditions such as temperature and humidity. Live values are compared with cinnamon-specific target ranges. Machine learning models (e.g., Random Forest) learn from historical data to recommend appropriate fertilizer usage and identify deficiencies, converting them into actionable recommendations.

Post-harvest "Quality Guard" - Multi-sensor devices monitor key environmental parameters inside warehouses: temperature, humidity, moisture, CO_2 levels, VOC levels, light intensity, motion detection, and air quality. Baseline ranges are used to evaluate conditions, and machine learning analytics detect mold and aroma-related risks early, triggering corrective guidance such as reducing humidity or controlling environmental exposure, while maintaining logs and trends.

Budget Analysis and Cost Management - The system allows users to track expenses, analyze costs, and manage financial aspects of cinnamon cultivation and storage. It provides insights into cost distribution and supports better financial decision-making.

Environmental Monitoring Component - This module monitors environmental conditions in related contexts and detects abnormal patterns. It provides additional alerts and insights that support overall system performance and decision-making.

Data Flow - Sensors / User Inputs → Firebase (Cloud Database) → Machine Learning Analysis → Mobile Dashboard & Alerts → Farm / Warehouse Actions.

Value - The platform delivers improved resource management, proactive post-harvest quality control, efficient cost management, and early risk detection, enhancing productivity, sustainability, and overall performance across the cinnamon value chain.

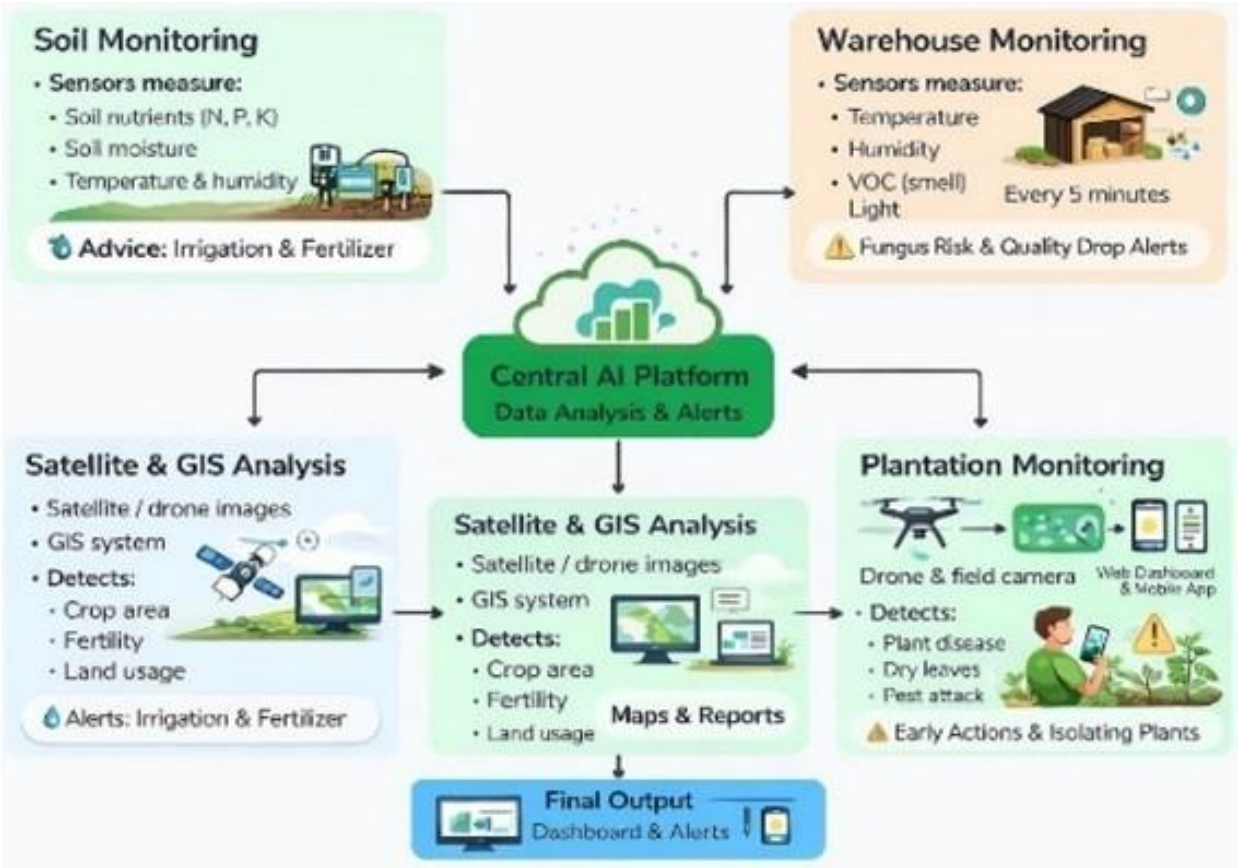

Figure 5: Overall System Diagram of SmartCinnamon

3.1 System Overview

The SmartCinnamon system is an integrated AI and IoT-based platform designed to support efficient monitoring and management across the cinnamon value chain, with a primary focus on post-harvest warehouse conditions. The system collects real-time environmental data using IoT sensors, processes the data through cloud-based services and machine learning models, and delivers insights, alerts, and recommendations to users through a mobile application.
At the core of the system is the mobile application, which acts as the main interface for users to monitor and interact with the system. The application is connected to a cloud-based platform (Firebase), where all sensor data is stored, managed, and synchronized in real time. This allows users to access up-to-date information about environmental conditions from any location.

The system consists of several interconnected components that work together to provide a complete monitoring solution. The warehouse monitoring component, also known as the Quality Guard, plays a central role in maintaining post-harvest cinnamon quality. It continuously monitors key environmental parameters including temperature, humidity, moisture, CO_2 levels, VOC levels, light intensity, motion detection, and overall air quality. These parameters are captured using multiple sensors and transmitted to the cloud for processing.

Machine learning techniques, specifically the Random Forest algorithm, are used to analyze the collected data and identify patterns that indicate potential risks. Unlike traditional monitoring systems, which rely on fixed threshold values, the SmartCinnamon system evaluates multiple environmental factors together to detect risks such as mold formation and aroma degradation at an early stage. This enables proactive monitoring and reduces the likelihood of quality loss.

When abnormal conditions are detected, the system generates real-time alerts and notifications through the mobile application. In addition to alerts, the system provides recommended actions to help users respond effectively to environmental changes. These features improve decision-making and reduce the need for manual monitoring.

The overall system follows a structured data flow: sensor data is collected and transmitted to

Firebase, where it is stored and processed. Machine learning models analyze the data and generate insights, which are then delivered to the mobile application in the form of dashboards, alerts, and recommendations. Users can monitor conditions and take appropriate actions based on this information.

Overall, the SmartCinnamon system provides a comprehensive and intelligent solution for monitoring warehouse conditions. By integrating IoT, machine learning, and mobile technologies, the system enhances storage quality, reduces post-harvest losses, and supports efficient and data-driven decision-making.

2. METHODOLOGY

2.1 Methodology

2.2 Commercialization Aspects of the Product

2.3 Testing and Implementation

3. RESULTS AND DISCUSSIONS

3.1 Results

3.2 Research Findings

3.3 Future Work

4. CONCLUSION

5. REFERENCES

[1] J. Smith, "Challenges in Post-Harvest Management of Cinnamon," *Agricultural Sciences Journal,* vol. 5, p. 110–115, 2021.